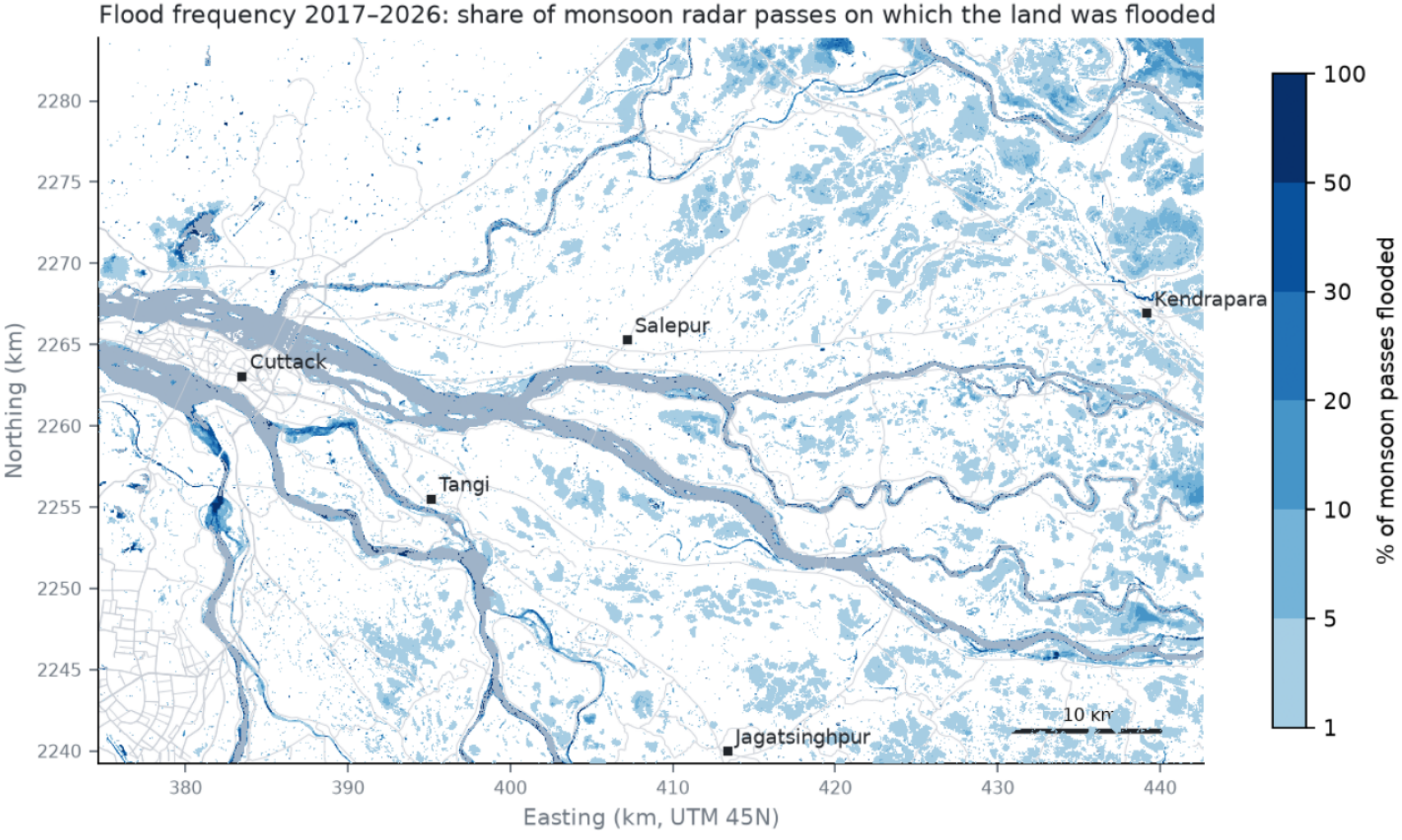
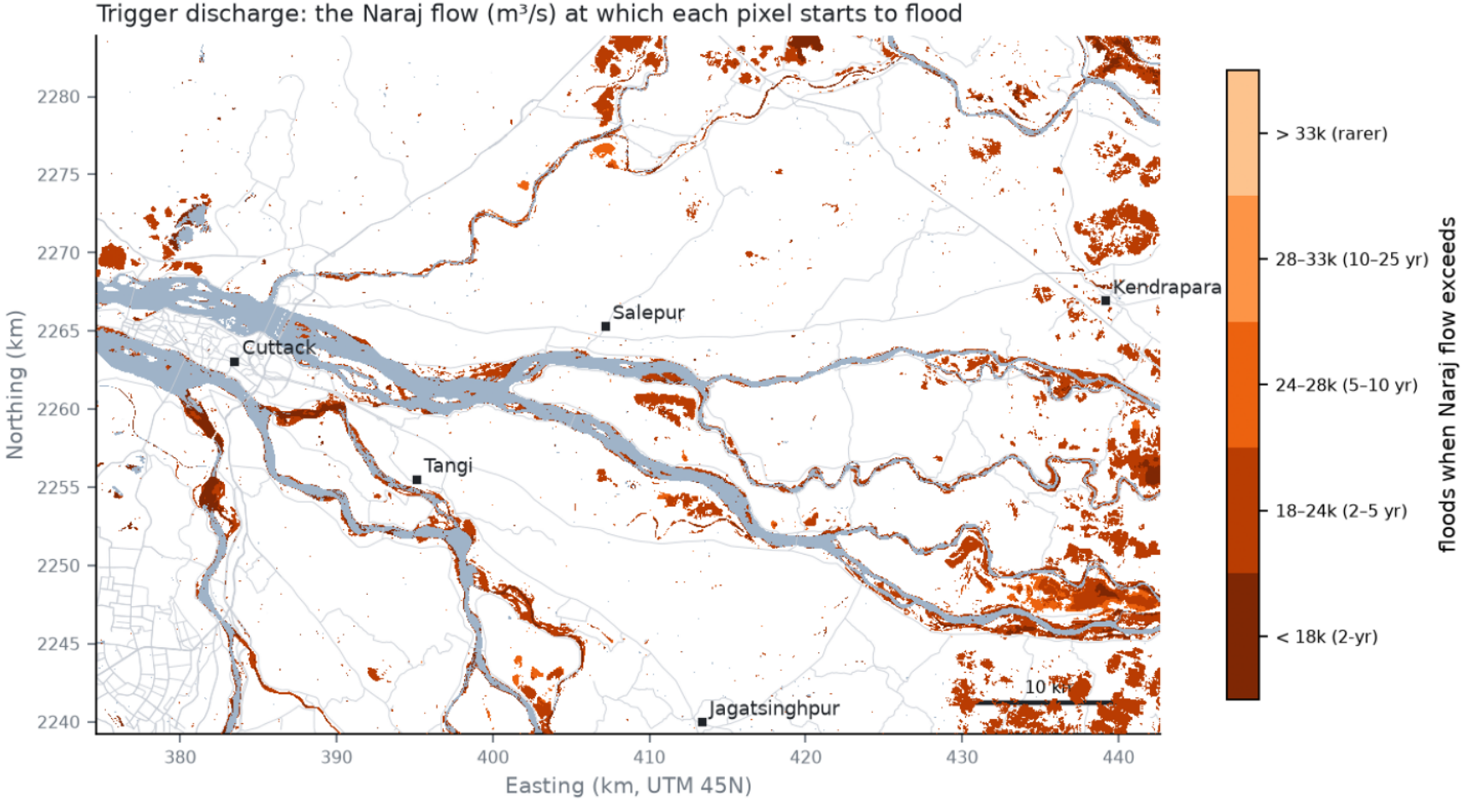
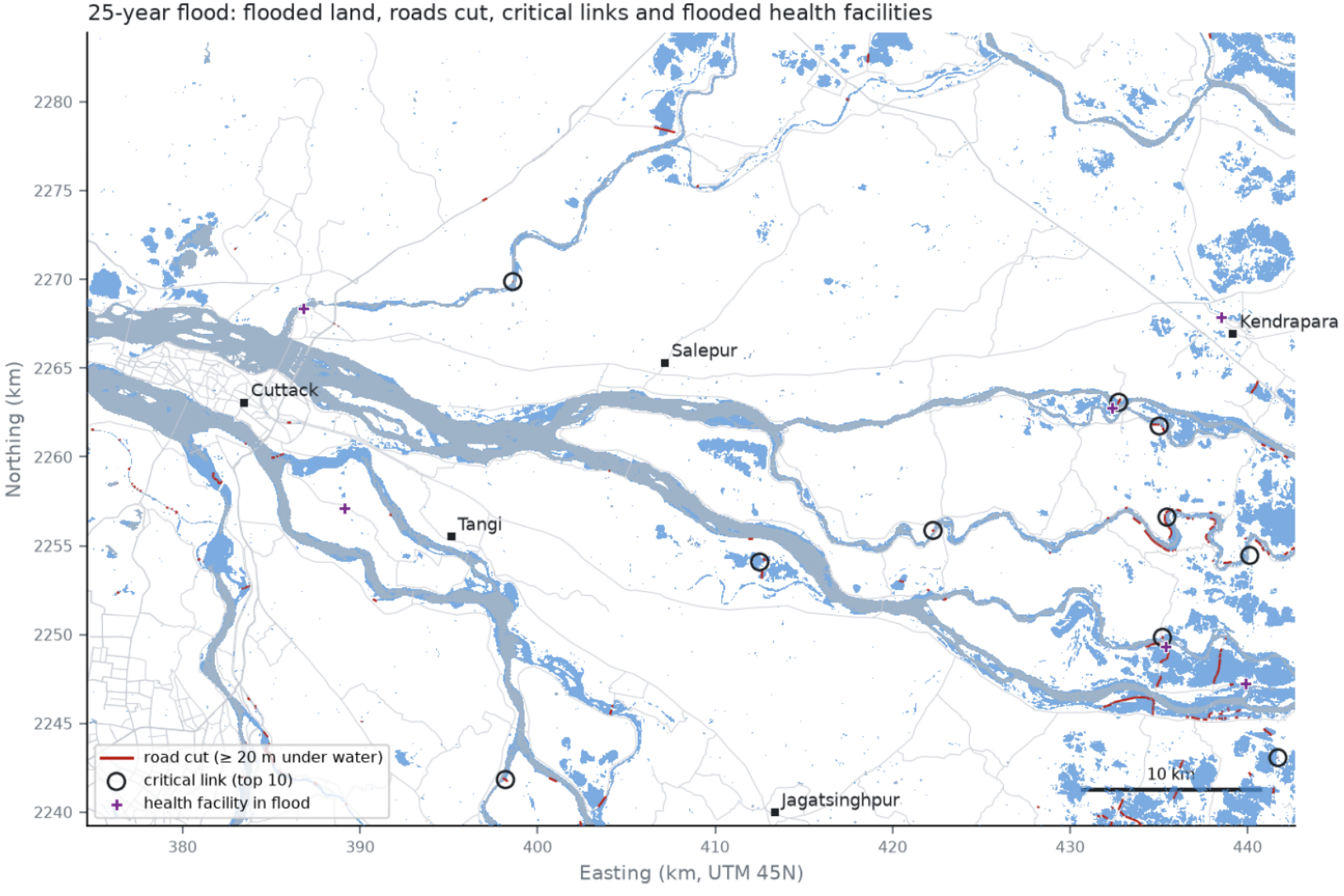
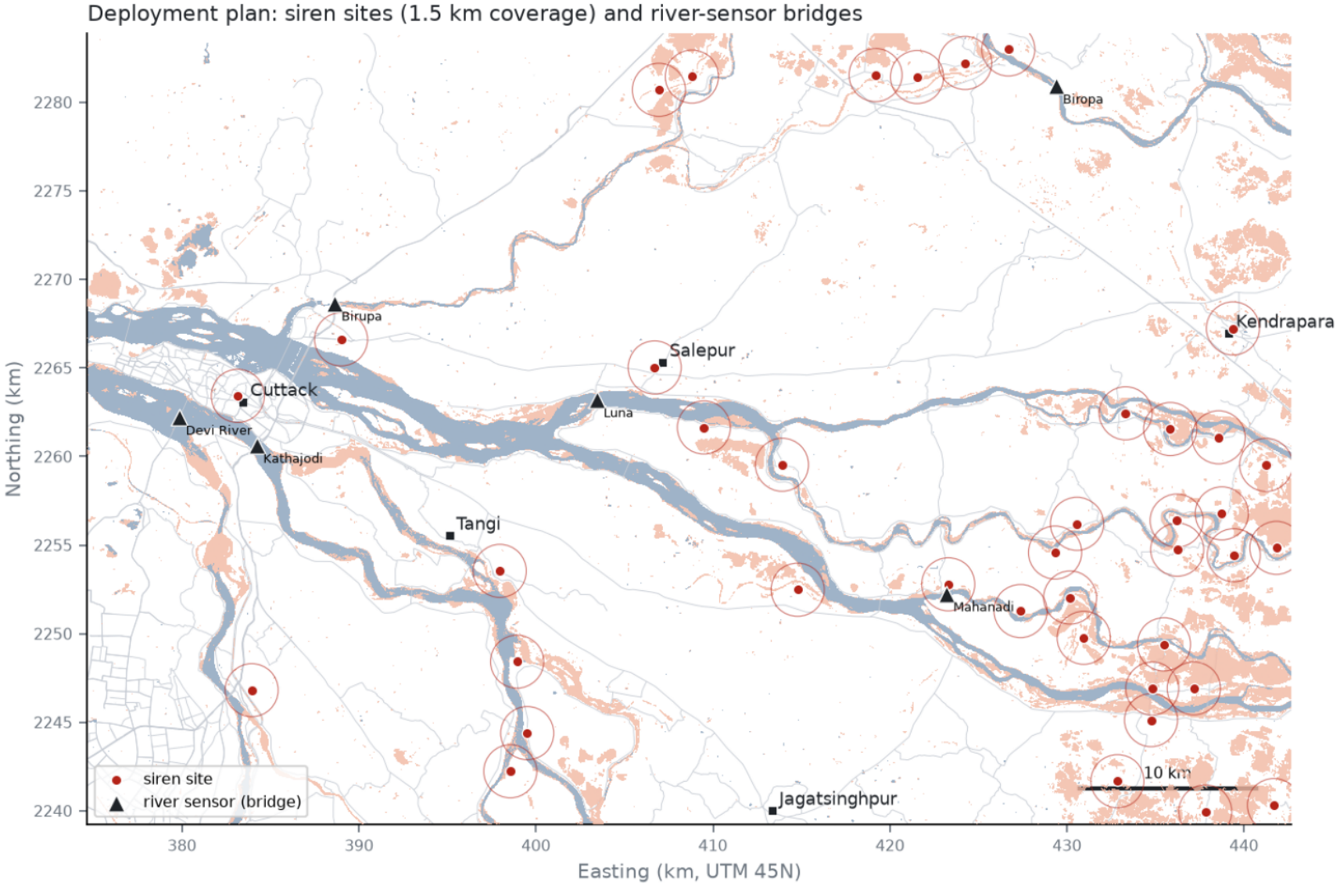
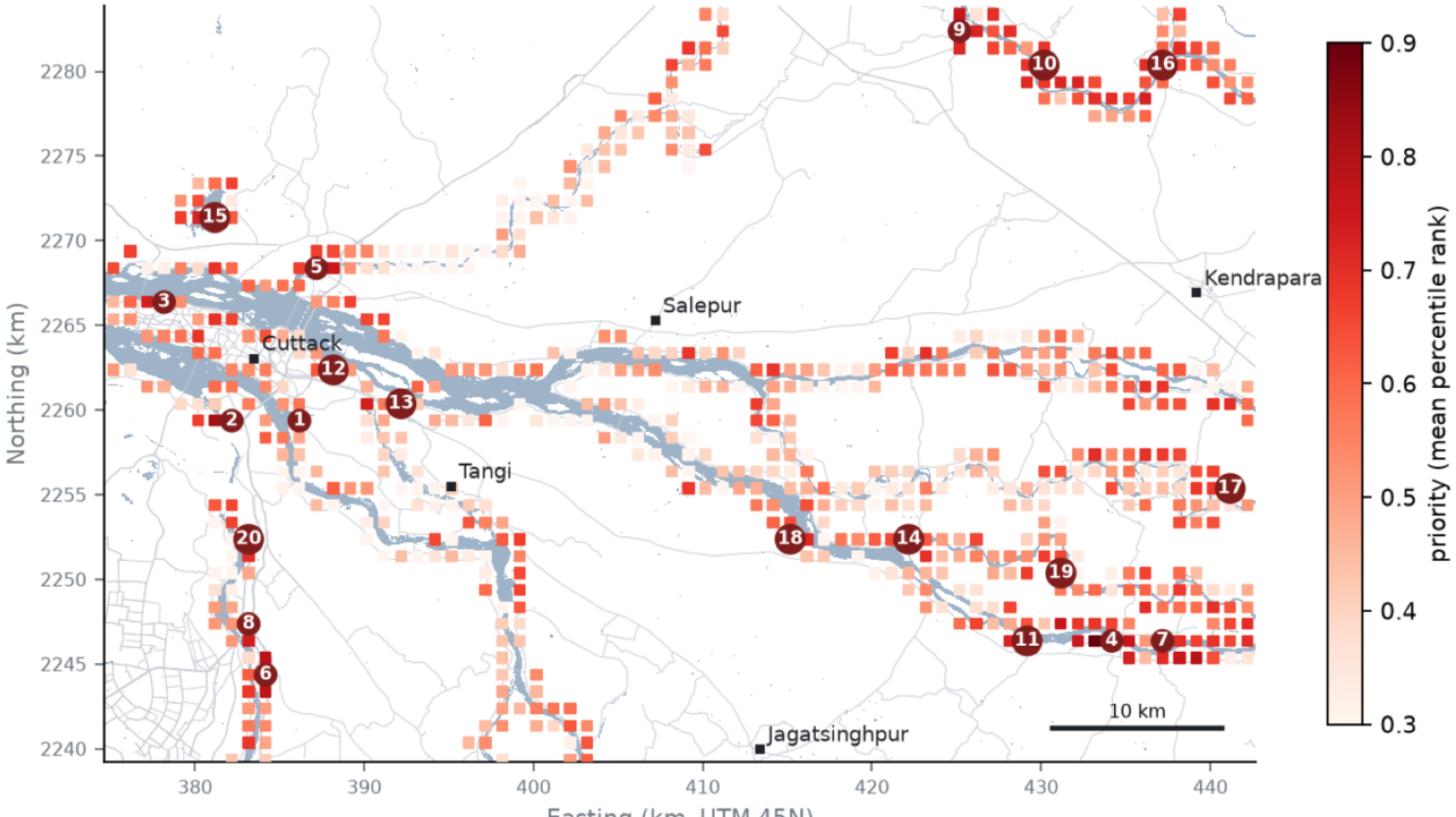

Flood frequency 2017–2026: share of monsoon radar passes on which the land was flooded
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
100
50
30
20
10
5
1
% of monsoon passes flooded


Trigger discharge: the Naraj flow (m³/s) at which each pixel starts to flood
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
> 33k (rarer)
28–33k (10–25 yr)
24–28k (5–10 yr)
18–24k (2–5 yr)
< 18k (2-yr)
floods when Naraj flow exceeds


25-year flood: flooded land, roads cut, critical links and flooded health facilities
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
road cut (≥ 20 m under water)
critical link (top 10)
health facility in flood
10 km


Deployment plan: siren sites (1.5 km coverage) and river-sensor bridges
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
Biropa
Kendrapara
Birupa
Salepur
Cuttack
Luna
Devi River
Kathajodi
Tangi
Mahanadi
Jagatsinghpur
siren site
river sensor (bridge)
10 km


River-bank inspection priority (1 km cells; field list = top 20 at least 3 km apart)
Northing (km)
Easting (km, UTM 45N)
priority (mean percentile rank)
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GT01	ground truth	20.485944	86.009344	atlas: never flooded; ask in near Jagatpur (8.1 km)
GT02	ground truth	20.350128	86.24033	atlas: never flooded; ask in near Nemalo (10.9 km)
GT03	ground truth	20.571656	86.019729	atlas: never flooded; ask in near Tangi (3.3 km)
GT04	ground truth	20.35512	86.409888	atlas: never flooded; ask in near Patkura (3.5 km)
GT05	ground truth	20.261454	86.44931	atlas: never flooded; ask in near Patkura (14.9 km)
GT06	ground truth	20.280172	86.425112	atlas: never flooded; ask in near Patkura (11.9 km)
GT07	ground truth	20.267712	85.898328	atlas: never flooded; ask in near Balikanthia (1.9 km)
GT08	ground truth	20.488688	86.271678	atlas: never flooded; ask in near Derabish (6.4 km)
GT09	ground truth	20.386076	86.425678	atlas: never flooded; ask in near Patkura (1.5 km)
GT10	ground truth	20.57739	86.043294	atlas: never flooded; ask in near Gokulpur (1.9 km)
GT11	ground truth	20.577155	86.390809	atlas: rarely (<10 %); ask in Nikirai
GT12	ground truth	20.489987	85.989565	atlas: rarely (<10 %); ask in near Jagatpur (5.3 km)
GT13	ground truth	20.603575	86.107807	atlas: rarely (<10 %); ask in near Bhattasagar Square (4.4 km)
GT14	ground truth	20.510507	86.106813	atlas: rarely (<10 %); ask in near Salepur (3.4 km)
GT15	ground truth	20.650277	86.279742	atlas: rarely (<10 %); ask in near Balachandrapur (6.1 km)
GT16	ground truth	20.310902	86.159109	atlas: rarely (<10 %); ask in near Jagatsinghpur (6.5 km)
GT17	ground truth	20.362829	86.295648	atlas: rarely (<10 %); ask in near Patkura (12.1 km)
GT18	ground truth	20.351348	85.893713	atlas: rarely (<10 %); ask in near Balikanthia (7.9 km)
GT19	ground truth	20.387079	86.071304	atlas: rarely (<10 %); ask in near Kishannagar (4.5 km)
GT20	ground truth	20.27329	86.368447	atlas: rarely (<10 %); ask in near Patkura (14.0 km)
GT21	ground truth	20.317229	86.375166	atlas: sometimes (10-30 %); ask in near Patkura (8.5 km)
GT22	ground truth	20.537002	86.026856	atlas: sometimes (10-30 %); ask in near Tangi (4.1 km)
GT23	ground truth	20.626241	86.324005	atlas: sometimes (10-30 %); ask in near Keutkandia (6.2 km)
GT24	ground truth	20.581768	85.987043	atlas: sometimes (10-30 %); ask in near Tangi (2.8 km)
GT25	ground truth	20.543733	86.254726	atlas: sometimes (10-30 %); ask in near Derabish (5.3 km)
GT26	ground truth	20.250658	85.945934	atlas: sometimes (10-30 %); ask in near Bhingarapur (2.5 km)
GT27	ground truth	20.507848	86.416211	atlas: sometimes (10-30 %); ask in Fakirabad
GT28	ground truth	20.291299	86.037018	atlas: sometimes (10-30 %); ask in near Gobindpur (4.1 km)
GT29	ground truth	20.463606	86.054163	atlas: sometimes (10-30 %); ask in near Salepur (6.0 km)
GT30	ground truth	20.42446	86.174021	atlas: sometimes (10-30 %); ask in Nemalo
GT31	ground truth	20.391823	86.315641	atlas: often (>=30 %); ask in near Patkura (8.7 km)
GT32	ground truth	20.270432	86.054573	atlas: often (>=30 %); ask in near Gobindpur (8.0 km)

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GT33	ground truth	20.284243	85.912192	atlas: often (>=30 %); ask in Balikanthia
GT34	ground truth	20.409974	86.385335	atlas: often (>=30 %); ask in near Patkura (2.4 km)
GT35	ground truth	20.407007	86.364453	atlas: often (>=30 %); ask in near Patkura (4.7 km)
GT36	ground truth	20.367938	86.263814	atlas: often (>=30 %); ask in near Nemalo (11.6 km)
GT37	ground truth	20.641705	86.008531	atlas: often (>=30 %); ask in near Bhattasagar Square (6.0 km)
GT38	ground truth	20.469996	86.173395	atlas: often (>=30 %); ask in near Nischintakoili (2.3 km)
GT39	ground truth	20.401244	86.276881	atlas: often (>=30 %); ask in near Nemalo (10.8 km)
GT40	ground truth	20.602502	85.919546	atlas: often (>=30 %); ask in near Choudwar (7.6 km)
SR01	siren site	20.50289	86.41865	rank 1: Fakirabad, covers 1,561 more people
SR02	siren site	20.31901	86.37564	rank 2: near Patkura (8.5 km), covers 1,067 more people
SR03	siren site	20.3872	86.41953	rank 3: near Patkura (1.5 km), covers 696 more people
SR04	siren site	20.33115	86.03182	rank 4: near Gobindpur (1.1 km), covers 602 more people
SR05	siren site	20.63049	86.24689	rank 5: near Balachandrapur (1.4 km), covers 565 more people
SR06	siren site	20.40471	86.38856	rank 6: near Patkura (2.4 km), covers 565 more people
SR07	siren site	20.63012	86.12477	rank 7: near Bhattasagar Square (7.0 km), covers 522 more people
SR08	siren site	20.45136	86.38491	rank 8: near Fakirabad (6.7 km), covers 486 more people
SR09	siren site	20.63768	86.27285	rank 9: near Balachandrapur (3.8 km), covers 481 more people
SR10	siren site	20.34158	86.38187	rank 10: near Patkura (5.9 km), covers 469 more people
SR11	siren site	20.34478	86.33846	rank 11: near Patkura (8.6 km), covers 393 more people
SR12	siren site	20.46579	85.87971	rank 12: Badambadi, covers 378 more people
SR13	siren site	20.40844	86.41295	rank 13: near Patkura (2.2 km), covers 366 more people
SR14	siren site	20.39099	86.44256	rank 14: near Patkura (3.9 km), covers 364 more people
SR15	siren site	20.25983	86.44169	rank 15: near Patkura (14.9 km), covers 279 more people
RS01	river sensor bridge	20.45458	85.84814	Devi River bridge, 2823 m, approaches dry in the 25-year flood: yes
RS02	river sensor bridge	20.44052	85.89012	Kathajodi bridge, 869 m, approaches dry in the 25-year flood: yes
RS03	river sensor bridge	20.51335	85.932	Birupa bridge, 543 m, approaches dry in the 25-year flood: yes
RS04	river sensor bridge	20.46498	86.07432	Luna bridge, 776 m, approaches dry in the 25-year flood: no
RS05	river sensor bridge	20.36684	86.26424	Mahanadi bridge, 458 m, approaches dry in the 25-year flood: yes
RS06	river sensor bridge	20.62609	86.32244	Biropa bridge, 424 m, approaches dry in the 25-year flood: no
GB01	bank inspection / soil sa	20.429	85.909	rank 1: behind-bank flood freq 4%, clay loam
GB02	bank inspection / soil sa	20.429	85.871	rank 2: behind-bank flood freq 6%, clay loam
GB03	bank inspection / soil sa	20.492	85.832	rank 3: behind-bank flood freq 2%, clay loam

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GB04	bank inspection / soil sa	20.314	86.369	rank 4: behind-bank flood freq 8%, clay loam
GB05	bank inspection / soil sa	20.511	85.918	rank 5: behind-bank flood freq 1%, clay loam
GB06	bank inspection / soil sa	20.294	85.891	rank 6: behind-bank flood freq 2%, clay loam
GB07	bank inspection / soil sa	20.314	86.398	rank 7: behind-bank flood freq 1%, clay loam
GB08	bank inspection / soil sa	20.321	85.881	rank 8: behind-bank flood freq 1%, clay loam
GB09	bank inspection / soil sa	20.639	86.282	rank 9: behind-bank flood freq 4%, clay loam
GB10	bank inspection / soil sa	20.621	86.33	rank 10: behind-bank flood freq 3%, clay loam
GB11	bank inspection / soil sa	20.314	86.322	rank 11: behind-bank flood freq 6%, clay loam
GB12	bank inspection / soil sa	20.457	85.928	rank 12: behind-bank flood freq 1%, clay loam
RL01	critical road link	20.38301	86.16167	unclassified: 10,351 people's hospital route, 1.03 km flooded (T25)
RL02	critical road link	20.27157	86.02512	unclassified: 9,421 people's hospital route, 0.34 km flooded (T25)
RL03	critical road link	20.45309	86.37705	tertiary: 8,261 people's hospital route, 0.71 km flooded (T25)
RL04	critical road link	20.39945	86.25511	unclassified: 7,657 people's hospital route, 0.14 km flooded (T25)
RL05	critical road link	20.34554	86.37918	tertiary: 7,045 people's hospital route, 0.14 km flooded (T25)
RL06	critical road link	20.40675	86.38139	tertiary: 6,371 people's hospital route, 2.06 km flooded (T25)
RL07	critical road link	20.46508	86.35512	unclassified: 6,210 people's hospital route, 0.34 km flooded (T25)
RL08	critical road link	20.28453	86.44175	residential: 6,171 people's hospital route, 0.03 km flooded (T25)
RL09	critical road link	20.38738	86.42615	tertiary: 4,863 people's hospital route, 0.31 km flooded (T25)
RL10	critical road link	20.52496	86.02748	unclassified: 4,638 people's hospital route, 0.05 km flooded (T25)
WL01	well levels (ground adv	20.3029	86.3751	near Patkura (10.2 km): 33% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL02	well levels (ground adv	20.6188	86.242	Balachandrapur: 28% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL03	well levels (ground adv	20.4029	86.3345	near Patkura (7.6 km): 33% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL04	well levels (ground adv	20.4113	86.3926	near Patkura (2.7 km): 60% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL05	well levels (ground adv	20.3901	86.4057	Patkura: 57% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL06	well levels (ground adv	20.4333	86.4369	near Patkura (5.8 km): 63% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL07	well levels (ground adv	20.4511	86.1323	near Salepur (4.4 km): 40% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL08	well levels (ground adv	20.4594	86.3606	near Fakirabad (7.7 km): 74% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
HF01	flooded health facility	20.40932	85.93741	PHC, Bentkar
HF02	flooded health facility	20.32195	86.42415	PHC, Hansura
HF03	flooded health facility	20.46205	86.35172	Government hospital, Nanakar
HF04	flooded health facility	20.50871	86.41009	Bhumika Hospital
HF05	flooded health facility	20.34104	86.38086	Sub Centre

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
HF06	flooded health facility	20.51068	85.9146	Unique Hospital